

AJJAY ADHITHYA V

Chennai, Tamil Nadu, India

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering; **CGPA: 9.06/10**

Chennai, India

2022 – 2026

Velammal Vidyalaya

CBSE All India Senior School Certificate Examination (AISSCE) - Class XII; **Grade: 95.2%**

Chennai, India

2021 – 2022

Experience

Centre for Cyber Physical Systems (CCPS), VIT Chennai [↗](#)

May 2025 – Aug 2025

Summer Research Intern

Chennai, India

- Conducted research on "Trade-Offs in AI-Text Detection," performing a comparative study of lightweight Transformer models including DistilBERT, ALBERT, and TinyBERT.
- Designed and implemented experiments to evaluate efficiency vs. performance trade-offs in detecting AI-generated text.
- Analyzed experimental data with faculty mentors, resulted in an IEEE conference publication (ICSCN 2025).

InstiBuzz - Campus Brand of IIT Madras [↗](#)

Aug 2024 – Oct 2024

Technical Team Intern - Full Stack

Chennai, India

- Contributed to the maintenance and optimization of the company website, ensuring smooth performance and scalability.
- Enhanced User Experience (UX) by implementing responsive design improvements and interactive features.
- Collaborated with the technical team to execute bug fixes, feature updates, and performance tuning.

Publications

LoRaWAN-Based Blood Sugar Monitoring to Enhance Diabetes Management in Rural Areas [↗](#)

Published in IEEE CICT 2025 | DOI: 10.1109/CICT67193.2025.11399013

Trade-Offs in AI-Text Detection: A Comparative Study of DistilBERT, ALBERT, and TinyBERT [↗](#)

Published in IEEE ICSCN 2025 | DOI: 10.1109/ICSCN67106.2025.11308419

Transforming Road Infrastructure Management: A Multimodal Road Quality Rating System [↗](#)

Published in IEEE ICCMC 2025 | DOI: 10.1109/ICCMC65190.2025.11140874

Projects

Automated Disaster Damage Assessment | Deep Learning, PyTorch, U-Net, ResNet-34 Aug 2025 – Nov 2025

- Built a unified deep learning model (U-Net with ResNet-34 encoder) for simultaneous building localization and multi-class damage classification from satellite imagery.
- Designed a custom CombinedLoss function with masking, achieving 0.97+ pixel accuracy and improving efficiency over traditional two-stage approaches.

Adaptive Traffic Signal Optimization [↗](#) | Reinforcement Learning, Python, AI

Sep 2024 – Nov 2024

- Developed a reinforcement learning-based traffic control system for adaptive signal optimization in urban environments.
- Simulated real-world traffic scenarios to reduce congestion and waiting time through dynamic signal timing adjustments.

Technical Skills

Machine Learning & AI: Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Model Optimization, Transfer Learning

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, OpenCV

Programming: Python (Primary), C/C++, Java, SQL, JavaScript

Web Development: React.js, Node.js, Express.js, HTML/CSS

Tools & Platforms: Linux, Git, GitHub, AWS, Docker